Robert Lee

Senior Software Engineer — Microsoft Azure AI Knowledge

hello@robertkl.com | robertkl.com | linkedin.com/in/robert-k-lee | github.com/robertklee

SUMMARY

Senior Software Engineer on Microsoft Azure AI Search, building the retrieval that grounds enterprise AI in the right knowledge at billion-vector scale. Technical lead for a team of six engineers and scientists building a new vector-search diversity capability. Experience spans vector search, agentic and information retrieval, RAG, and distributed systems.

TECHNICAL SKILLS

Languages: C++, C#, Java, Python, TypeScript
Search & AI: Vector search, HNSW, quantization, information retrieval, hybrid search, agentic retrieval, RAG, computer vision
Systems: Distributed and durable systems, graph databases, performance engineering, SIMD, authentication, production diagnostics

EXPERIENCE

Senior Software Engineer **Mar. 2025 – Present**
Microsoft — Azure AI Search C++, C#, Java, Python *Redmond, WA*

- **Technical lead for a team of six engineers and scientists** building a new **vector-search diversity capability**. Set the technical direction and own delivery end to end, from leadership buy-in and algorithm analysis through distributed architecture, cross-team alignment, and hands-on implementation.
- Bridged research and production: distilled the strengths of each stage of the research algorithm, then rethought its architecture for **distributed execution** with targeted algorithmic and systems innovations that preserve them at production scale. The capability reduces homogeneous, redundant nearest-neighbour results for grounding, e-commerce, and recommendations.
- Led research-to-production delivery of **agentic-retrieval filter and boost generation**. Reformulated an unbounded filter-synthesis problem into a **bounded, verifiable operator set**, used production analysis to show it covered real customer workloads, and brought research and product teams to consensus.
- Architected a benchmarking system from scratch to profile highly variable agentic-retrieval workloads across CPU, memory, and throughput, modeling latency distributions and tool-calling iterations; resolved memory optimizations it surfaced, and proposed and shipped the **production billing model** for agentic retrieval in a new serverless enterprise search offering.
- Integrated retrieval into agent workflows, grounding LLM agents in governed enterprise knowledge across tool calling, multi-agent orchestration, and RAG. Mentor and onboard engineers, lead design reviews, and drive architecture decisions.

Software Engineer II **Jun. 2022 – Feb. 2025**
Microsoft — Azure AI Search C++, C#, Java, Python *Redmond, WA*

- Helped build and scale **vector search and quantization** for broad production adoption, driving vector storage, quantization, and relevance across billions of embeddings.
- Drove **vector quantization from Public Preview through GA**, now widely adopted, through binary vectors, scalar and binary quantization, and SIMD-accelerated distance computation. Reported customer results: **8–32x cost savings** and **up to 20x lower latency**, varying by workload.
- Optimized vector-distance kernels with **SIMD, loop unrolling, independent vector accumulators, and fused multiply-add**, breaking serial dependencies to expose instruction-level parallelism.
- Strengthened relevance with hybrid-search subscore fusion and score thresholding across blended vector and keyword retrieval.
- Designed and shipped data-driven **HNSW quota enforcement** tied to physical resource utilization, **cutting limit overshoot by 100x** through cross-team design collaboration.
- Owned an **extensible facet-aggregation engine** from spec through delivery: a lexer, parser, and evaluator using BNF grammar, shunting-yard parsing, and Reverse Polish Notation, backed by extensive A/B test coverage.
- Built a vector-engine test suite that caught a critical quantization bug before release; root-caused cross-team production incidents and reviewed distributed-systems and vector-algorithm changes.

Software Engineer **Jul. 2021 – May 2022**
Microsoft — Azure AI Search *Redmond, WA*

- Delivered index aliases to Public Preview, letting applications switch indexes with **no code changes or downtime**.
- Modernized a core telemetry-database table through a phased, zero-downtime migration that preserved data integrity and delivered a **50–100x query speedup**.

EARLIER MICROSOFT EXPERIENCE

- **Software Engineer Intern, Azure Cognitive Search** (Sep.–Dec. 2020): designed and shipped the team's second most requested API and backend feature in C#, released under a preview API.
- **Software Engineer Intern, Azure Search** (Jun.–Aug. 2019): built a TypeScript search-website generator with query suggestions and filtering, plus a portal JSON editor.
- **Software Developer Intern, Microsoft Garage** (Jan.–Apr. 2018): built a cross-platform C# mobile app for offline chest X-ray classification, with its image pipeline, CI, iOS share extension, and TensorFlow Android integration.

SELECTED PROJECTS

Human Pose Estimation Using Deep Neural Networks [Code] [Demo] *Jan. 2021 – May 2021*

- Led a student team training a heatmap-based 2D pose model on COCO 2017 **from scratch with random weight initialization**; architected the cloud training pipeline, model, and augmentation, and led visualization and deployment.

Semantic Road Segmentation Using Convolutional Neural Networks [Code] *Jul. 2020 – Sep. 2020*

- Trained a U-Net CNN on KITTI Road using Keras, achieving **up to 99.1% F1** and 91% in the worst case.
- Researched the architecture from computer-vision literature; built the training pipeline, custom losses, augmentation, and Colab training; fixed decoder bugs and addressed shadow and occlusion edge cases.

Self-Supervised Monocular Depth Estimation [Code] *Jan. 2020 – May 2020*

- Built a limited Monodepth2 variant in TensorFlow on ~**136,000 DrivingStereo image pairs** without depth labels: a ResNet-18 U-Net with stereo reconstruction, edge-aware smoothness, multi-scale loss, auto-masking, and minimum reprojection.

Battlesnake Reinforcement Learning Agent *Jan. 2019 – May 2019*

- Trained a keras-rl agent via self-play and public snakes to survive real-time Battlesnake games against up to seven opponents.

LEADERSHIP & COMMUNITY

Industry Mentor, SENG 321 — Requirements Engineering **2024; 2026 – Present**
University of Victoria *Victoria, BC*

- Since Aug. 2026: authored an open-ended challenge from unsolved problems in production AI; coach a student team through requirements elicitation, stakeholder analysis, scoping, and design rationale to a clickable prototype.
- Sep.–Dec. 2024: one of 18 mentors for a 120-student cohort; authored an LLM-chatbot reliability and safety challenge and served as the client stakeholder for requirements elicitation.

Founder & Program Director, Seniors' Digital Literacy Program **Jul. 2015 – Oct. 2021**
University of Victoria *Victoria, BC*

- Founded and directed the program for six years, growing it to **180 volunteers and 650+ seniors served across 30 workshops**; secured departmental sponsorship, funding, lab space, and administrative support.
- Designed concurrent beginner and advanced streams, built a volunteer pipeline with the UVic CS Course Union and two high schools, and handed the program to successors who sustained it.

Chair & Vice-Chair, IEEE Student Branch **Sep. 2019 – Jan. 2021**
University of Victoria *Victoria, BC*

- Co-led a six-person executive and 70-member branch; built a **14-workshop series reaching 350+ students**, about five times the membership, across Git, machine learning, circuit design, and hardware prototyping.
- Authored much of the curriculum and taught Git, machine learning, and circuit design; secured $1,000 and lab access, recruited alumni and industry instructors, and made the series recurring.

Founder & Conference Chair, Tech & Business Strategy Conference **Aug. 2019 – Nov. 2019**
University of Victoria *Victoria, BC*

- Conceived and secured university buy-in for a **200+ attendee conference**; built and led the organizing committee, managed logistics, recruited industry speakers, and emceed the event.

EDUCATION

Bachelor of Engineering, Electrical & Computer Engineering **Sep. 2016 – Apr. 2021**
University of Victoria Computational Intelligence Specialization *Victoria, BC*

- **97% cumulative average**; **2nd-highest CGPA** in the university's graduating class; Dean's List. Earned **20+ scholarships and awards worth over $100,000**.
- Jamie Cassels Undergraduate Research Award: researched hardware acceleration for machine-learning neural networks.

SELECTED AWARDS & CREDENTIALS

- **Schulich Leader Scholarship** (2016) — $80,000 from Canada's largest undergraduate STEM scholarship program; **one of 25 engineering recipients nationwide** in a 50-person cohort chosen from 1,500+ school nominees.
- **Y Combinator AI Startup School — Inaugural Cohort** (Jun. 2025) — **~8% selected from 30,000 applicants** for a two-day San Francisco conference for 2,500 technical students, researchers, and builders.
- **National Champion, Michael Smith Science Challenge** (2014) — **1st in Canada among 1,753 competitors** with a record-setting 97.5% score, against a 42.5% national average.
- **1st Place, Western Engineering Competition — Senior Design** (2020) — built a robot to collect Martian artifacts, against 50 competitors from 11 other Western Canadian engineering universities.
- **Jamie Cassels Undergraduate Research Award** (2019); **1st Place (3x), UVEC Engineering Competition — Senior Design** (2017–2019); **Governor General's Academic Medal, Bronze** (2016).